Elisha Afari

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PROFESSIONAL SUMMARY

A highly analytical and motivated recent graduate in Materials Engineering, blending a robust foundational understanding of physical sciences with rapidly evolving expertise in Data Science and Advanced Computing. My academic and project experience is defined by a proactive pursuit of interdisciplinary excellence, translating complex engineering challenges into data-driven solutions. I possess a strong technical core in Machine Learning (ML) and am actively building proficiency in Data Engineering principles to manage the full data lifecycle. This is complemented by practical development experience in Web and Mobile Application Development. Crucially, I excel in high-fidelity communication and professional presentation, leveraging expert-level command of the Microsoft Office Suite (Excel, PowerPoint, Word) to ensure technical insights are clear, compelling, and accessible to diverse stakeholders seeking to contribute to research or projects at the intersection of materials science, data optimization, and advanced analytics, leveraging a unique fusion of engineering intuition and computational mastery.

SKILLS

- Machine Learning
- Data Analysis and Visualization
- Data Engineering
- Microsoft Excel: Pivot tables, macros, dashboards, advanced analytics
- Website Development
- Programming: MATLAB, Python
- Mobile Application Development
- Software Development
- Strong organizational, leadership, and problem-solving skills
- Languages: English (Bilingual), French (Beginner), Asante Twi (Native),

EXPERIENCE

Welding and Fabrication — MAC Mining and Construction Partners (September 2024 – October 2024)
Research & Teaching Assistant — KNUST Materials Engineering Department (2025 – Present)

EDUCATION

- BSc. Materials Engineering — Kwame Nkrumah University of Science and Technology, 2025
- High School, General Science — Swedru Senior High School, 2021

RESEARCH PROJECTS

- **Machine learning based predictions of perovskite solar cells stability**
- **A Hybrid Machine Learning and SCAPS-1D Simulation**

• Built, trained and optimized machine learning and deep learning models to predict perovskite stability and efficiency.
• Ran and Automated simulations using python scripts
• Analyzed perovskite solar cells fabrication processes and parameters to optimize stability and efficiency

AWARDS

- **Most improved Materials Engineering Student, 2024** — awarded for outstanding academic performance and improvement.

LEADERSHIP POSITIONS

- **President — Materials Innovation Hub KNUST Obuasi Campus** • Directed activities for 100+ members and organized mentorship programs.

• Organized skill trainings.

■ **Deputy Head of Academic Affairs — Materials Engineering Dept, KNUST Obuasi, 2024 & 2025**
• Organized tutorials and peer-learning sessions to support struggling students.
• Improved average class performance by 20% through structured study initiatives.

REFERENCES

■ Emmanuel Acheampong Tsiwah – KNUST, Lecturer | ea.tsiwah@knust.edu.gh |+233541886057
■ Martin Luther Yeboah — KNUST, Lecturer | lm.yeboah1.coe@gmail.com | +233 24 020 2148